

Solid Oxide Electrolyzer Cells for Industrial Decarbonization

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Introduction

Due to global warming and the world political instability the search for alternative fuels has been made more urgent than ever. As societies continue to heavily rely on fossil fuels for electrical generation, transport sector, and industrial production, greenhouse gas (GHG) emissions keep increasing and their impacts have become harder to mitigate. This has created a strong need for energy carriers that have a low carbon intensity while still being practical, scalable and preferably compatible with existing infrastructure or requires minimal additions. Among the energy carriers being considered, hydrogen stands out as one of the most promising due to its ability to address two major challenges at once: clean energy storage and clean fuel supply.

Hydrogen makes a suitable fuel because it is a way to both store and deliver energy. It can be made using renewable electricity and electrolysis, or even heat waste and stored for later energy use. That makes it especially useful for balancing intermittent generation methods, especially solar power. Hydrogen can also help in sectors that are difficult to electrify directly, such as steel production, aviation, and heavy-duty trucking. When hydrogen is used in a fuel cell or combusted as a fuel, it can provide energy with little or no direct carbon emissions. That dual role makes hydrogen a particularly strong candidate in the broader effort to move away from fossil fuels.

Solid Oxide Electrolysis Cells (SOECs) are a promising type of high-temperature fuel cell that use electricity to split water into hydrogen gas. They run in a regenerative mode, or in reverse operation of a Solid Oxide Fuel Cell (SOFC). SOECs are particularly compelling due to their ability to transform both thermal and electrical energy into chemical energy, enabling clean and efficient hydrogen production that can be run entirely off of solar energy or waste heat (Chen et al., 2025).

Aim

The paper has three key aims that can be summarised as the following:

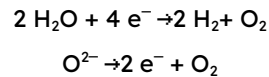
- To summarize the fundamental mechanics of SOECs and explain the operation of the unique features of co-electrolysis and reversible SOEC (rSOEC) configurations.
- To assess the integration of SOEC systems into industrial applications such as Power-to-Hydrogen and make comparisons to conventional electrolysis methods.

- To identify key challenges and opportunities for improving SOEC performance and its role in industrial decarbonization and future hydrogen production.

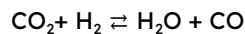
System Description

Overview

Solid oxide electrolyzer cells (SOECs) are typically operated between 650 °C and 900 °C, consist of two porous electrodes separated by a dense pure oxide-ion conducting electrolyte, and can achieve technical efficiencies up to 100 % due to advantageous thermodynamics and reaction kinetics. In steam electrolysis, steam (H₂O) is electrochemically reduced to hydrogen (H₂) at the negatively charged fuel electrode. The resulting oxide ions migrate through the electrolyte to the positively charged oxygen electrode, where they are oxidized to molecular oxygen as:



The operating conditions and equilibrium process impact the (reverse) water gas shift reaction ((R)WGS) as expressed in the equation:



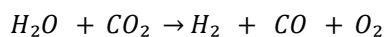
occurring during the co-electrolysis process, with the product ratio on the fuel side being variable. Syngas and methane are derived from methanation with an H₂:CO ratio of 3:1. The high operating temperature is the main advantage of SOECs, which results in lower ohmic losses compared to other electrolyzers as well as advantageous kinetics and thermodynamics. The process enthalpy ΔH of the required endothermic steam reduction process can be supplied in the form of heat and electrical energy (Gibbs energy ΔG). With increasing temperatures, the enthalpy increases slightly for steam reduction while the electrical energy demand ΔG becomes smaller at higher temperatures. The remaining energy demand can be supplied in the form of heat.

In addition, the internal cell resistances lead to the production of Joule heat, which adds to the enthalpy for reactant-to-product conversion. At the thermo-neutral voltage, the exact power required for the endothermic electrolysis process is produced, and therefore no external thermal management is necessary other than from compensating for the heat loss to the surroundings. The energy efficiency of up to 100 % and the use of inexpensive catalyst materials make SOEC systems more cost-effective in comparison to alkaline and PEM electrolyzer

Advanced Concepts

Co-electrolysis

The electroreduction of two or more reactants simultaneously is referred to as co-electrolysis, and has gained traction in recent years with water and carbon dioxide feedstocks, for the production of syngas and reduction of greenhouse gases using the following equation (Zheng, 2017).



High temperature co-electrolysis (HTCE) in SOEC systems come with a number of advantages compared to comparable low temperature electrolysis. On the one hand there are lower cost, energy consumption and higher energy efficiency, due to using a singular reactor in a unified process (Zheng, 2017). On the other hand, the high operational temperature of the SOEC provides improved thermodynamics and faster reaction kinetics (Wolf, 2022).

The ability of SOECs to efficiently electrolyze steam and CO₂ adds pathways for PtM and PtL integration, since the produced syngas can be used in downstream Fischer-Tropsch Synthesis (FTS) reactions with the additional benefit of adjustable H₂/CO ratios, due to the free choice of feedstock ratios (Ferrete, 2025). With this system-level coupling, the high temperatures also create optimal catalysis conditions, for said downstream FTS reactions of the on site produced syngas (Becker, 2012).

Reversible Operation (rSOEC)

Reversible solid oxide electrolysis cell (rSOEC) refers to the inverse operation of the electrolysis cell in the form of a solid oxide fuel cell (SOFC), producing extra electricity for the stack operation based on energy availability and system needs. This is of special interest for carbon free operations that rely solely on waste heat and renewable energy sources (RES) like solar and wind. Due to the inconsistency of their power output their integration can lead to problems, such as fluctuations in the operating temperature causing faster degradation of the solid electrolyte and electrodes. Thus, the system can run a number of cells in fuel cell mode, parallel to the electrolyzer mode, to make up for any power shortcomings during periods of limited energy availability (Ferrete, 2025)

Additionally, the reversible operation allows the entire rSOEC system to function as a grid-scale energy storage, where during excess renewable generation the cells run electrolysis mode, chemically storing energy, and then switch to fuel cell mode and feeding electricity back into the grid during low RES output or high energy demand (Banasiak, 2023).

Application Pathways & Industrial Integration

High Temperature Waste Heat Industries

SOEC provides a solution for industry decarbonization by both utilizing high-temperature waste heat and cogenerating hydrogen for use as a carbon-emitting substitute in industrial burners. Industrial heat waste can be integrated in electrolysis operation for steam production by way of an evaporator and increase system efficiency. Furthermore, hydrogen blending is becoming a common practice that utilizes existing natural gas infrastructure to lower fuel consumption and operational costs in addition to smoothly transitioning to cleaner energy systems. The hydrogen produced by SOECs can be recycled into such a hydrogen blend, resulting in a closed loop that significantly increases system efficiency. According to simulations by Banasiak et al., this type of thermal coupling can increase the overall efficiency by 15%

(Banasiak et al., 2023; Pastore et al., 2025). As hydrogen increasingly replaces fossil fuels in these industries in the future, the economic viability of hydrogen cogeneration using SOECs will grow.

TABLE 1: Comparison of maximum continuous waste heat temperature of various industries.

	Cement	Glass	Refractory	Steel
Maximum Waste Heat Temp. [°C]	450	600	1200	1400
Continuous Waste Heat Temp. [°C]	300	380-500	180-300	110-140

The essential indicator of an optimal industry for SOEC integration is an available heat source that is both continuous and within the allowed operating range due to the degrading effect of heat cycling. Industries that can couple well with SOECs include iron and steel, cement, glass and refractory processing. These industries are particularly suited for integration due to their increasing utilization of hydrogen. However, while waste heat is provided constantly by these industries, the use of batch processing results in fluctuating temperatures which can lead to fast degradation. The values collected by Banasiak et al. shown in **Table 1** are the available continuous waste heat by relevant sectors above 100°C which can be utilized by SOECs (Banasiak et al., 2023).

Nuclear technology, particularly Small Modular Reactors (SMRs) also presents an opportunity to integrate electricity generation by partially supplying thermal energy for high temperature electrolysis. Vargas-Ferrer et al. predicts a reduction of hydrogen cost by up to 14% as compared to low-temperature electrolyzers (Vargas-Ferrer et al., 2026).

Long Term Energy Storage

Moreover, the reversible nature of the SOEC is a key function which extends the use case of the cell stack to provide heat or electricity as needed. This functionality becomes important in industrial processes where waste heat is already directed to district heating; SOECs can utilize more waste heat during summer months when demand for district heating is low, and then operate in fuel cell mode during winter months to provide heating in place of heat pumps or gas boilers, thus improving the decarbonization of the district heating sector (Banasiak et al., 2023).

Application Pathways

One of the main strengths of SOECs is their versatility, which allows for a wide range of energy conversions. The most important applications can be seen below in **FIG. 1**, which represents the industrial applications for SOEC mentioned in various papers from 2015 to 2025. As can be seen, the classical utilization was mainly hydrogen and methane production, which expanded to methanol, electricity and lastly power to liquid (PtL) biofuel production (Ferrete, 2025).

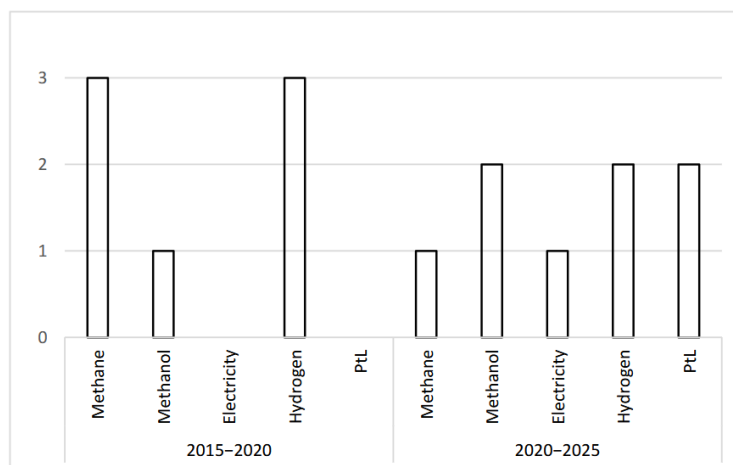
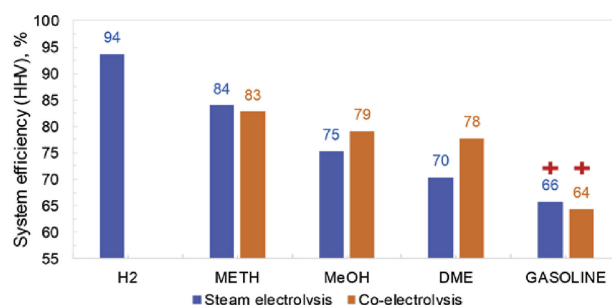


FIGURE 1: Number of publications discussing industrial applications of SOEC



(a) HHV system efficiency for the considered process alternatives, with LHV efficiency given in SI. The red cross stands for the possibility of a further efficiency increase by using waste heat available.

FIGURE 2: HHV system efficiency of various Power-to-X applications of SOECs. Also compares benefits of co-electrolysis functionality (Wang et al., 2019).

FIG 2., produced by Wang et al., shows the system efficiencies of producing methane, methanol, dimethyl ether, and gasoline using SOECs. Notably, there is a significant increase in the benefit of co-electrolysis as the produced molecules become more complex, especially for DME. Additionally, the inclusion of waste heat in the process will boost system efficiencies, especially for gasoline production (Wang et al., 2019).

Key Performance Indicators

Three notable Key Performance Indicators (KPIs) for SOECs are (1) electricity consumption and heat demand under nominal conditions, (2) average current density, and (3) reversible capacity. A table from the Clean Hydrogen Joint Undertaking indicating these target KPIs for 2030 can be used for comparison to the current state of SOECs.

TABLE 2: KPI comparison for SOECs (Clean Hydrogen Joint Undertaking, 2022).

No	Parameter	Unit	State of Technology	Target
			2020	2030
1	Electricity consumption	kWh/kg H ₂	40	37
	@ nominal capacity			
7	Heat demand	A/cm ²	9.9	8
	@ nominal capacity			
9	Current density		0.6	1.5
	Reversible capacity	%	25	40

When examining KPI-1, Zhao et al. reported an electricity consumption of 3.77 kWh/Nm³-H₂, or 33.7 kWh/kg H₂ (Note: the value used for comparison by Ferrete et al. did not account for this conversion, so it is incorrect), which is already below the target of 37 kWh/kg H₂. Similarly, their reported heat demand of 5.45 kWh/kg H₂ is significantly below the 2030 target of 8 kWh/kg H₂. This shows demonstrated improvements compared to other electrolyzers in terms of power-to-hydrogen efficiency (Zhao et al., 2021; Ferrete et al., 2025).

When examining KPI-7, which compares the average current density, Chen et al. estimates a range of 0-2.5 A/cm², or an average of 1.25 A/cm², which indicates a reasonable expectation for SOECs to reach the target of 1.5 A/cm² by 2030 (Chen et al., 2025).

KPI-9 measures the reversible capacity of SOECs, or the ratio of the nominal rated power in fuel cell mode to the electrical power when in electrolyzer mode. One recent example is of the reversible cell from Ficili et al. with a reversible capacity of 33.33%, indicating a reasonable expectation for SOECs to reach the target of 40% by 2030 (Ficili et al., 2022).

Comparative Assessment

TABLE 3: Comparisons of current main electrolysis methods.

	AEC	PEMEC	SOEC
Operating Temperature [°C]	60-80	50-80	650-1000
CAPEX [€/kW]	220-500	220-530*	590-1400*
Current Density [A cm ⁻²]	0.2-0.4		0-2.5**
Lifetime [kilohours]	50-130*	50-130*	10-90*
Maturity	Mature	Commercial	Demonstration

* (Buhler, 2024)

** (Chen et al., 2025)

Currently, three main types of H₂O electrolysis cells are used in hydrogen production systems: alkaline electrolysis cells (AECs), proton exchange membrane electrolysis cells (PEMECs), and solid oxide electrolysis cells (SOECs). AECs, known for their stability and durability, have been widely used in the industry for decades, but their low current density can limit their efficiency and scalability in the future (Chen et al., 2025). PEMECs can be coupled with renewable energy, but the high cost of platinum catalysts restrict its economic viability in large-scale applications (Chen et al., 2025). In comparison, SOECs have no requirement for costly catalysts, have a high energy conversion efficiency potential (approaching 100%), can harness waste thermal energy from other systems and have the flexibility of switching between a fuel cell mode and electrolysis mode (Chen et al., 2025).

Unlike conventional electrolysis cells, SOECs operate at significantly higher temperatures which allows for the use of a wider range of feedstocks, including renewable sources and thermal waste (Pande, 2023). This includes many industries which produce a high amount of waste heat such as the steel and concrete industries.

The operating temperature of AECs and PEMECs are limited to 80C, which limits the efficiency of concentrated solar collectors as a power source compared to SOECs (Schmidt et al., 2017). Solar energy is an ideal feedstock as it can supply electrical energy through photovoltaics, as well as high-temperature heat through solar heat collectors (Chen et al., 2025).

Challenges & Constraints

Material Degradation

The high temperature nature of SOECs—which run between 700 and 900°C—is a quality that differentiates it from conventional electrolyzers and enables it to be flexible in thermal feedstock. However, this operational condition presents its biggest challenge as well. Any variation in temperature, especially start-ups and shutdowns, create nonuniform thermal gradients which can lead to thermal stress and fatigue in the cell materials (Chen et al., 2025). This challenge is significant because the benefit of SOECs is dependent on waste heat sources and renewable energy sources, which often fluctuate in temperature. SOECs take anywhere from 16 to 24 hours to transition from operation to cold standby (Fabiani et al., 2025). Maximum temperature gradients have been identified at around 8-10K/cm to avoid these long-term damages (Fabiani et al., 2025).

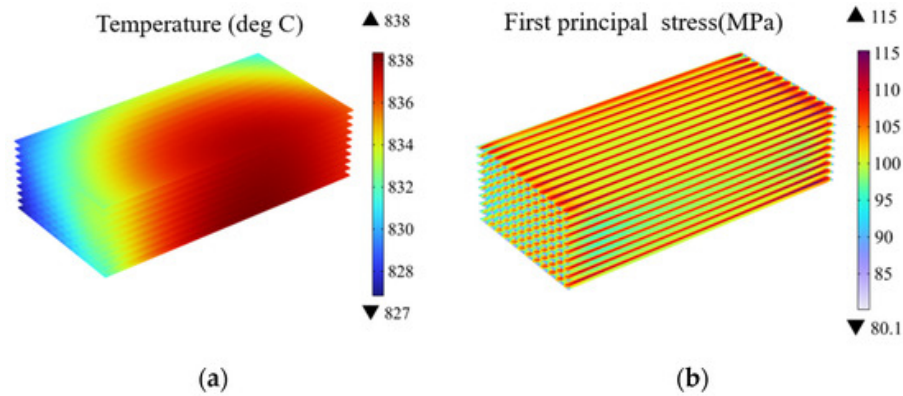


FIGURE 3: Effect of minor fluctuations of temperature in a range of 10°C to stress on the material (Wang et al., 2023).

Liquid CO₂ energy storage (LCES) can act as a regulator to address intermittent power sources, capable of shaving RES production peaks and generating electricity (Ferrete et al., 2025). During periods of surplus energy, CO₂ is compressed, cooled and stored at a lower temperature and high pressure. When electricity is required, the CO₂ is heated and expanded in a turbine.

Local Overpotentials

Local overpotentials, which directly result in degradation of the cell, can be caused by excessively high or low oxygen activity (Chen et al., 2025). This results in O₂ bubbles forming inside of the solid electrolyte or delamination (Chen et al., 2025). These overpotentials can also be caused by increases in current density (Chen et al., 2025).

Short Operation Times

One of the most critical challenges that prevents SOECs from becoming mainstream is its inability to perform for long durations. However, the only way to mitigate the damage-inducing temperature gradients is to operate SOECs at a continuous temperature for tens of thousands of hours (to be competitive with conventional electrolyzers). It is possible to operate SOECs at lower current densities, which can reduce degradation and extend the lifespan of the cells; however, this reduces the rate of electrolysis, or, while in Fuel Cell mode, reduces the power output. When the current density is reduced sufficiently to maintain a reasonable degradation rate, the benefits of efficiency are effectively nullified in comparison to conventional AECs. For instance, Hu et al. demonstrated an exceptional current density of 5 A/cm² but only managed an operation time of 95 hours (Hu et al., . However, Smitshuysen et al. demonstrated a successful long-term operation of over 20,000 hours in a 70-cell stack using AC:DC technology at only 0.5 A/cm² (Smitshuysen et al., 2025). This showcases the primary challenge in SOEC technology of balancing current density and operational lifetime that would make its utilization economically viable.

Thermal management

In order to maintain a stable temperature to achieve long-term operation using unsteady heat sources such as renewables or waste heat, the following methods are used:

- A sweep gas (primarily air) flows through the oxygen electrode, which stabilizes temperature gradients, displaces hydrogen in the case of a leakage, and lowers the partial pressure of oxygen allowing for greater efficiency (Fabiani et al., 2025).
- The cell is operated at a lower voltage (in endothermic mode), which reduces current density and can extend its lifespan at the cost of a lower hydrogen production rate (Fabiani et al., 2025).

System costs

SOECs have significantly higher system costs than conventional methods. According to Jiménez-Martín et al., the Levelized Cost of Hydrogen (LCOH) was estimated to be $6.45 \text{ €}\cdot\text{kg}^{-1}$, compared to the current cost of grey and blue hydrogen of $1.35 \text{ €}\cdot\text{kg}^{-1}$ and $2.75 \text{ €}\cdot\text{kg}^{-1}$, respectively (Jiménez-Martín et al., 2024). The majority of the LCOH is in Operational Expenditures (OPEX), primarily through electricity costs and stack degradation and replacement (Jiménez-Martín et al., 2024).

Strategies, Design Recommendations & Future Role

Efficiency

The efficiency advantage of solid oxide electrolysis cells arises from their high operating temperature, which alters the thermodynamic distribution between electrical and thermal energy inputs for steam electrolysis.

Without continuous heat integration for steam generation and stack temperature control, SOEC systems increasingly rely on electrical heating, diminishing their efficiency advantage relative to low-temperature electrolyzers (Flis and Wakim, 2023; Wu et al., 2024).

Designing and deploying SOECs primarily in heat-rich environments where high-grade thermal energy is continuously available, such as steel, glass, and synthetic fuel production industries. Operation close to thermoneutral conditions minimizes electrical consumption while limiting excessive internal heat generation, which would otherwise increase thermal stress and accelerate material degradation (Ferrete et al., 2025; Banasiak et al., 2023).

Flexibility

Solid oxide electrolysis cells exhibit inherently limited tolerance to rapid load variations and frequent start-stop operation due to their elevated operating temperature and ceramic construction. Spatial temperature gradients induced by fast transients generate thermo-mechanical stresses that accelerate electrode delamination, electrolyte cracking, and long-term performance degradation (Pan et al., 2026; Kim-Lohsoontorn et al., 2023).

Dynamic operation studies indicate that gradual load adjustments, rather than rapid ramping, enable SOEC stacks to accommodate variations in electrical input while maintaining acceptable temperature stability and degradation rates (He et al., 2025; Zhu et al., 2025). Advanced control strategies, including model-predictive and load-path optimisation methods, have been shown to significantly reduce internal

temperature gradients and improve dynamic performance compared to conventional control approaches (Allan et al., 2024).

As mentioned earlier, rSOC operation allows for a mode-switching capability which enables long-duration energy storage and demand-responsive operation without requiring additional electrochemical hardware. This allows for system-level flexibility for applications such as industrial energy management, microgrids, and seasonal energy storage.

It is however critical to distinguish between energy balancing flexibility and fast grid-balancing services. Due to thermal inertia, SOEC and rSOC systems are not well suited for high-frequency regulation or rapid power fluctuations.

Durability

The performance of the SOEC is undoubtedly pivoted by the durability and the kinetics of the complicated electrochemical reactions including nickel coarsening in fuel electrodes, interfacial delamination and degradation at electrode-electrolyte interfaces. Degradation increases internal resistance resulting in progressively higher operating voltages and reduced efficiency (Sun et al., 2019; Wolf et al., 2022).

Recent studies emphasize the need for degradation-aware operation that favors moderate current densities and stable thermal conditions, thereby maximizing lifetime-averaged efficiency rather than beginning-of-life performance (Giridhara et al., 2024; Chen et al., 2025). In their work (Tong et al) emphasized the in-situ/operando approach for developing electrodes with unique structures and composition as a new strategy.

Conclusion

SOECs are one of the most promising electrolysis technologies for low-carbon hydrogen production because they can achieve very high efficiency if integrated with continuous high-grade heat sources such as industrial waste heat, solar heat or nuclear cogeneration. Their ability to produce hydrogen, co-electrolysis, and reversible operation also makes them particularly valuable for Power-to-X applications and broader industrial decarbonization.

At the same time, SOECs are still limited by high capital cost, temperature sensitivity, slow startup and shutdown, and shorter stack lifetime compared with mature low-temperature technologies. These constraints mean that SOECs are not yet the best choice for highly dynamic or standalone systems, but they are highly attractive in heat-rich environments where operating conditions can remain stable.

In conclusion the most promising pathway is therefore not broad adoption everywhere, but targeted integration in industries such as steel, cement, glass, and synthetic fuel production where waste heat and steady operation can be utilised. Better thermal management, reducing stack degradation, and improved reversible-system design, SOECs could become a key technology for efficient hydrogen production and industrial energy storage.

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